STEVENS INSTITUTE OF TECHNOLOGY

EE627A: Ensemble Modeling

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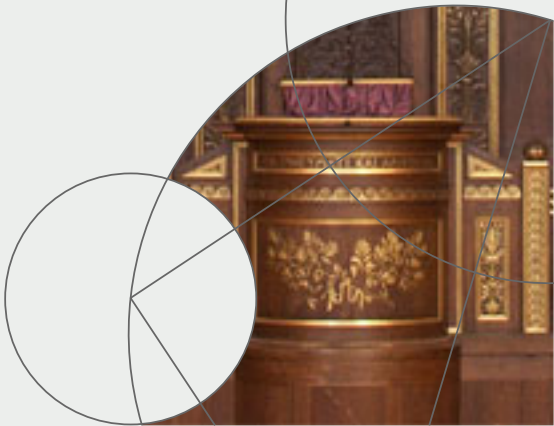


Review The Final Project

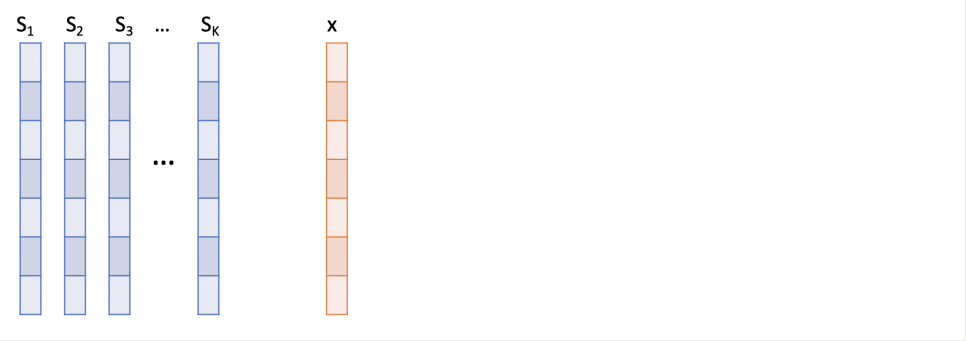
- You are asked to submit a vector which contains the binary sequence to predict which 3 tracks the test user will like and which 3 tracks will dislike.
- You have submitted various of algorithms for your final project. These submitted solutions are $\hat{\mathbf{s}}_1,\ \hat{\mathbf{s}}_2,\ \cdots,\ \hat{\mathbf{s}}_K$ and the ground truth is $\hat{\mathbf{x}}$.
- The solution vector contains half of elements as "0"s and half of "1"s.
- We do not like "0"s and prefer to change the binary sets $\{0,\ 1\}$ as $\{-1,\ +1\}$. Since the elements $\{0,\ 1\}$ are categorical data sets, this change will not change any categorical data distribution.
- Mathematically, we can easily use below formula to map $0 \to (-1)$ and $1 \to (+1)$.

$$\mathbf{x} = 2\hat{\mathbf{x}} - 1 \qquad \text{and} \qquad \mathbf{s}_i = 2\hat{\mathbf{s}}_i - 1, \quad i = 1, 2, \cdots, K$$

- We will explain why and how we can benefit from this non-trivial expression format change.

Exploit Multiple Models of the Same ML Task?

- Let us use the final project example to demonstrate the ensemble learning method.
- You have submitted various of algorithms for your final project. These submitted solutions are $\mathbf{s}_1, \ \mathbf{s}_2, \ \cdots, \ \mathbf{s}_K$ and the ground truth is $\mathbf{x}$.

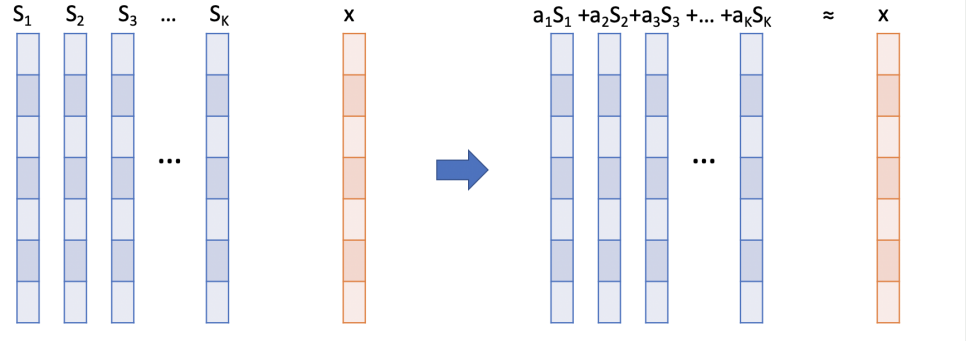


- The problem of interest is to take advantage of all different ML algorithms and combine them together to boost the overall performance.

Linear Ensembling

- We can build an optimized linear combination of all the predictions to approximate the ground truth solution.
- That is, given submitted solution vectors $\mathbf{s}_1,\ \mathbf{s}_2,\ \cdots,\ \mathbf{s}_K$, we look for a set of weights to combine a new vector which is close to the $\mathbf{x}$ as much as we can.

$$a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K \approx \mathbf{x}$$



Optimization Problem

- This ensembling question can be converted to a typical optimization problem, i.e., a minimum distance linear fitting problem.

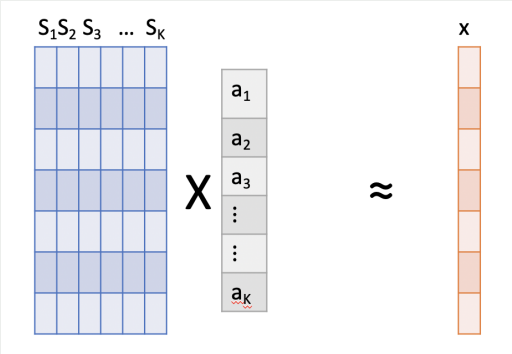
$$a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K \approx \mathbf{x}$$

$$\Rightarrow \qquad \arg\min_{a_1, a_2, \cdots, a_K} \|\mathbf{x} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K)\|^2$$

Optimization Problem

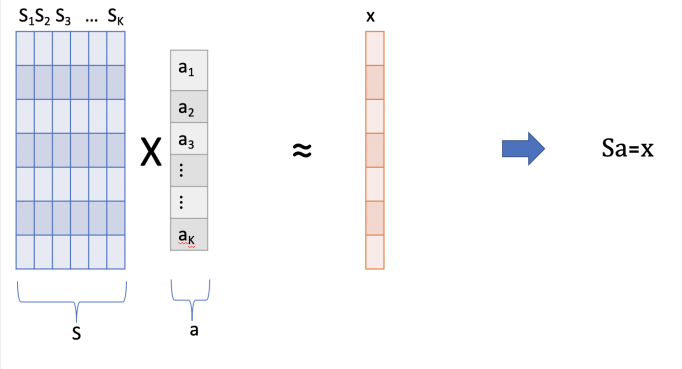
- To facilitate the derivation, we form the vector expression in a compact matrix form.
- Stack all the solution vectors as a matrix and the weights form a vector

$$a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K = [\mathbf{s}_1, \mathbf{s}_2, \cdots, \mathbf{s}_K] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_K \end{bmatrix}$$



Optimization Problem

- Denote a matrix $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \cdots, \mathbf{s}_K]$ and a column vector $\mathbf{a} = [a_1,\ a_2,\ \cdots,\ a_K]^T$.



- The optimization problem is reduced to an expression

$$\mathbf{Sa} \approx \mathbf{x} \qquad \Rightarrow \qquad \arg\min_{a_1, a_2, \cdots, a_K} \|\mathbf{x} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K)\|^2 = \arg\min_{\mathbf{a}} \|\mathbf{x} - \mathbf{Sa}\|^2$$

Least Squares Solution

- The minimum distance optimization problem has a classic solution: Least-Squares(LS) solution.

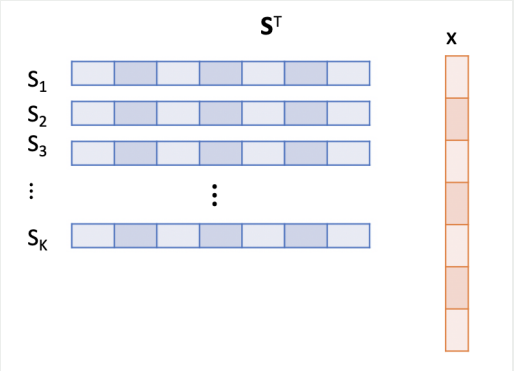
$$\arg\min_{\mathbf{a}} \|\mathbf{x} - \mathbf{S}\mathbf{a}\|^2 \qquad \Rightarrow \qquad \mathbf{a}_{\mathsf{LS}} = \left(\mathbf{S}^T\mathbf{S}\right)^{-1}\mathbf{S}^T\mathbf{x}$$

- The LS solution $\mathbf{a}_{\mathsf{LS}}$ tells us how to put weights on different solutions and generate a linearly combined new solution vector which should have the smallest distance to the ground truth vector $\mathbf{x}$.

☞ Observe the expression $\mathbf{a}_{\mathsf{LS}}$, we know the matrix $\mathbf{S}$ but not the ground truth $\mathbf{x}$ (otherwise, why do we bother doing this final project here).

What is $\mathbf{S}^T\mathbf{x}$?

- Let us examine the term $\mathbf{S}^T\mathbf{x}$ within the LS solution $\mathbf{a}_{\text{LS}}$.
- As shown in the below plot, $\mathbf{S}^T\mathbf{x} = \begin{bmatrix} \mathbf{s}_1^T\mathbf{x} \\ \mathbf{s}_2^T\mathbf{x} \\ \vdots \\ \mathbf{s}_K^T\mathbf{x} \end{bmatrix}$



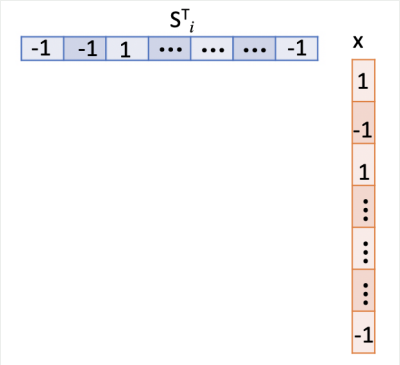
What is $\mathbf{s}_i^T$ in $\mathbf{S}^T\mathbf{x}$?

- Let us examine one specific term $\mathbf{s}_i^T\mathbf{x}$. It is essentially the sum of element-by-element multiplications.
- 4 types of multiplications.

  1. $1 \times 1 = 1$
  2. $1 \times (-1) = -1$
  3. $(-1) \times (-1) = 1$
  4. $(-1) \times 1 = -1$

☞ Remarks:

- Case 1 and 3 are predictions equal to the truth
- Case 2 and 4 are false predictions



Why Change "0"s to "-1"s with $\mathbf{s}_i$?

- Now we can explain why we changed the binary set $\{0, 1\}$ to $\{-1, 1\}$.
- From the previous multiplication discussions. If we keep the binary set $\{0, 1\}$, we have below 4 types of multiplications within $\mathbf{s}_i^T\mathbf{x}$

☞ Use the set $\{0, 1\}$

	Truth: 0	Truth: 1
Predition: 0	0× 0= 0	0× 1=0
Predition: 1	1× 0= 0	1× 1=1

☞ Use the set $\{-1, 1\}$

	Truth: -1	Truth: 1
Predition: -1	(-1)× (-1)= 1	(-1)× 1=-1
Predition: 1	1× (-1)= -1	1× 1=1

- One can see that, when using the set $\{-1, 1\}$, the mulitplication products "1" indicate the cases where the prediction and the truth matching, while the mulitplication products "(-1)" indicate the cases where the prediction and the truth not matching.

Why Change "0"s to "-1"s with $\mathbf{s}_i$?

- When the prediction is equal to the ground truth, you have the multiplication equal to "1", no matter the prediction is "1" or "0" itself.
- When the prediction is not matching the ground truth, you always have the multiplication equal to "-1".
- Overall, when $\mathbf{s}_i^T\mathbf{x}$ is essentially the sum of element-by-element multiplications, we will have

$$\mathbf{s}_i^T\mathbf{x} = \sum_{l=1}^{N} \mathbf{s}_i[l] \times \mathbf{x}[l]$$

$$= \text{number of succeed predictions} \times 1 + \text{number of failed predictions} \times (-1)$$

where N is the total predictions with the test user trackIDs.

What is $\mathbf{s}_i^T$ in $\mathbf{S}^T\mathbf{x}$?

- We reform the expression $\mathbf{s}_i^T\mathbf{x}$ as

$$
\begin{aligned}
\mathbf{s}_i^T\mathbf{x} &= \text{number of succeed predictions} \times 1 + \text{number of failed predictions} \times (-1) \\
&= \text{number of succeed predictions} \times 1 - \text{number of failed predictions} \times 1 \\
&= \text{number of succeed predictions} - (N - \text{number of succeed predictions}) \\
&= 2 \times (\text{number of succeed predictions}) - N \\
&= N\left[2 \times \frac{\text{number of succeed predictions}}{N} - 1\right] \\
&= N\left[2 \times (\text{correct rate}) - 1\right] \\
&= N(2P_i - 1) \qquad \text{Denote } P_i \triangleq \text{correct rate}
\end{aligned}
$$

- The $P_i \triangleq$ correct rate is the score you obtained when you submit your solution to the Kaggle webpage.

What is $\mathbf{S}^T\mathbf{x}$?

- From the previous discussion, we have

$$\mathbf{S}^T\mathbf{x} = \begin{bmatrix} \mathbf{s}_1^T\mathbf{x} \\ \mathbf{s}_2^T\mathbf{x} \\ \vdots \\ \mathbf{s}_K^T\mathbf{x} \end{bmatrix} = \begin{bmatrix} N(2P_1 - 1) \\ N(2P_2 - 1) \\ \vdots \\ N(2P_K - 1) \end{bmatrix}$$

where $P_1,\ P_2,\ \cdots,\ P_K$ are all your scores from the Kaggle submissions.

What is the Least Squares Solution $\mathbf{a}_{\mathsf{LS}}$?

- Let us revisit the Least Squares (LS) solution $\mathbf{a}_{\mathsf{LS}}$:

$$\arg\min_{\mathbf{a}} \|\mathbf{x} - \mathbf{S}\mathbf{a}\|^2 \qquad \Rightarrow \qquad \mathbf{a}_{\mathsf{LS}} = \left(\mathbf{S}^T\mathbf{S}\right)^{-1}\mathbf{S}^T\mathbf{x}$$

- $\mathbf{S} = [\mathbf{s}_1,\ \mathbf{s}_2,\ \cdots,\ \mathbf{s}_K]$ contains all the solution submissions at Kaggle, which are known. Therefore the calculation of $\left(\mathbf{S}^T\mathbf{S}\right)^{-1}$ is straightforward.

- $\mathbf{S}^T\mathbf{x}$ has been demonstrated to be reformed as

$$\mathbf{S}^T\mathbf{x} = \begin{bmatrix} N(2P_1 - 1) \\ N(2P_2 - 1) \\ \vdots \\ N(2P_K - 1) \end{bmatrix}$$

where $P_1,\ P_2,\ \cdots,\ P_K$ are all your scores from the Kaggle submissions.

The Best Overall Model: Ensembling Model

- Your various solution submissions and the corresponding evaluation responses from Kaggle can be utilized to form the ensembling model $\mathbf{s}_{\text{ensemble}}$.

$$\mathbf{s}_{\text{ensemble}} = a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \cdots + a_K\mathbf{s}_K = \mathbf{S}\cdot\mathbf{a}_{\text{LS}} = \mathbf{S}\left(\mathbf{S}^T\mathbf{S}\right)^{-1}\mathbf{S}^T\mathbf{x}$$

- $\mathbf{s}_{\text{ensemble}}$ is the linear combination, or weighted average of all the previous submissions, where each submission contains equally half binary results $\{-1, 1\}$.
- As the weights in $\mathbf{a}_{\text{LS}}$ are fractional numbers, the weighted average will not be integers $\{-1, 1\}$ any more.
- Recall that our goal is to predict two sets of classes for each test user: Like or Dislike.
- ☞ The last step is to sort every 6 scores of each user in $\mathbf{s}_{\text{ensemble}}$ and mark the top 3 as "1" and the rest 3 as "0".

$$\tilde{\mathbf{s}}_{\text{ensemble}} = \text{sort } \& \text{ threshold}\left\{\mathbf{s}_{\text{ensemble}}\right\}$$